

Ian Maywar

New York, NY • ijmaywar@gmail.com • +1 (585) 451-3356 • <https://www.ianmaywar.com>

Summary

Data scientist and quantitative researcher with experience building analytical workflows for large, complex datasets and developing statistical and machine learning models to extract actionable insights. Background in experimentation, time-series analysis, and geospatial data, with an ability to translate statistics and model outputs into clear narratives for stakeholders.

Experience

Research Associate (Data Scientist)

Feb 2025 – present

The City University of New York – New York, NY

- Designed Python-based ETL and analysis workflows for 300M+ global observations from the NASA GLOBE program, integrating satellite and in-situ sensor data to assess data quality, bias, and validity.
- Deployed statistical and machine learning methods including Gaussian Processes and time-series decompositions.
- Communicated insights and recommendations to NASA program leadership through technical reports, visualizations, and presentations to inform program strategy and data governance.

DEVELOP Researcher (Geospatial Data Analyst)

Jan 2025 – Apr 2025

NASA – Remote

- Led quantitative analysis of environmental and behavioral vulnerability in Chatham County, Georgia using Python and QGIS, combining satellite imagery, climate indicators, and demographic data.
- Designed composite indices and applied models to characterize population exposure patterns, delivering findings to the county's Resilience Program through briefings, presentations, and written reports.

Graduate Research Assistant (Data Scientist)

Aug 2022 – Dec 2024

Stony Brook University – Stony Brook, NY

- Modeled and analyzed 40 longitudinal environmental metrics in R and Python to assess ecosystem health in the New York Bight, producing a technical report for the New York State Department of Environmental Conservation.
- Analyzed 55,000+ hours of high-frequency GPS and accelerometer data using advanced statistical and machine learning methods to infer behavioral states and movement dynamics of seabirds.
- Authored publications and presentations and collaborated with interdisciplinary teams across academic fields.

Research Assistant (Lab Manager)

Sep 2021 – May 2022

Brown University – Providence, RI

- Developed and executed laboratory protocols for dietary analysis of Yellowstone National Park herbivores using DNA metabarcoding of fecal samples.
- Managed laboratory resources, curated research datasets, and led technical training for undergraduate researchers.

Education

Stony Brook University, Stony Brook, NY

Aug 2022 – Dec 2024

- Master of Science in Marine and Atmospheric Sciences, 3.96/4.00 GPA.

Skidmore College, Saratoga Springs, NY

Sep 2017 – May 2021

- Bachelor of Arts in Mathematics (honors), Bachelor of Arts in Biology (honors), *summa cum laude*, 3.91/4.00 GPA.

Skills

Programming Languages: Python (Pandas, NumPy, SciPy, Statsmodels, scikit-learn, Matplotlib, Seaborn, Cartopy), R (tidyverse, terra, ggplot2, lme4, mgcv), MATLAB (Statistics and Machine Learning Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox), SQL (Postgres), JavaScript (Google Earth Engine).

Software & Platforms: ArcGIS, QGIS, Git/GitHub, VS Code, JupyterLab, Colab, Microsoft Suite, Google Workspace.

Methods: Statistical modeling, machine learning, timeseries analysis, remote sensing, hypothesis testing, research design.